Day 43 MCQ

Question 1

Correct

Mark 1.00 out of 1.00

Consider the following three table to store student enrollements in different courses.
Student(EnrollNo, Name)
Course(CourseID, Name)
EnrollMents(EnrollNo, CourseID)
(EnrollNo,CourseID are primary keys)

What does the following query do?

```
SELECT S.Name
FROM Student S, Course C, Enrollments E
WHERE S.EnrollNo = E.EnrollNo AND
C.Name = "DBMS" AND
E.CourseID = C.CourseID AND
S.EnrollNo IN
(SELECT S2.EnrollNo
FROM Student S2, Course C2, Enrollments E2
WHERE S2.EnrollNo = E2.EnrollNo AND
E2.CourseID = C2.CourseID
C2.Name = "OS")
```

Select one or more:

- [x] a. Name of all students who are enrolled in "DBMS" and "OS"

Background Reading: The above query is an example of **nested query** i.e. query within a query. Firstly the inner query is solved and then the outer one depending on the result of the inner query.

WHERE IN returns values that matches values in a list or subquery. WHERE IN is a shorthand for multiple OR conditions. `Here, firstly the inner query is solved. It returns all the Enrollment Numbers (SELECT S2.EnrollNo) of students where the students' enrollment number matches with the enrollment number of the courses (WHERE S2.EnrollNo = E2.EnrollNo) which have the course IDs whose Course Name is "OS" (E2.CourseID = C2.CourseID and C2.Name = "OS").`

Hence all the enrollment IDs are filtered out for the students who are enrolled for the "OS" course.

`The outer query works similarly and filters out all the all tuples where the Students Enrollment Number matches with the Enrollment Number where the course ID's are for the course names "DBMS" (S.EnrollNo = E.EnrollNo AND C.Name ="DBMS" AND E.CourseID = C.CourseId) and additionally matches with the ones that are returned by the inner query i.e. Enrollment Number of students who are enrolled for the course "OS".`

Hence the above queries returns names of all students (SELECT S.Name) who have enrolled for both courses "DBMS" and "OS".
Hence option (B).

- [ ] b. Name of all students who are either enrolled in "DBMS" or "OS" or both
- [ ] c. Name of all students who are either enrolled in "DBMS" or "OS" courses
- [ ] d. None of the above

Your answer is correct.

The correct answer is: Name of all students who are enrolled in "DBMS" and "OS"

Question 2

Correct

Mark 1.00 out of 1.00

Consider the following Employee table

ID salary DeptName
1 10000 EC
2 40000 EC
3 30000 CS
4 40000 ME
5 50000 ME
6 60000 ME
7 70000 CS

How many rows are there in the result of following query?

```
SELECT E.ID
FROM Employee E
WHERE EXISTS (SELECT E2.salary
FROM Employee E2
WHERE E2.DeptName = 'CS'
AND E.salary > E2.salary);
```

Select one or more:

- a. 0
- b. 4
- c. 5 ✓

Here in the above question, there is a correlated subquery because the subquery references the enclosing query (relation Employee renamed as E)
The subquery (SELECT E2.salary FROM Employee E2
WHERE E2.DeptName = 'CS')
Filters out E2 relation as (all tuples where DeptName is CS and the respective salaries)
Now the correlated query works as follows:

```
SELECT E.ID
FROM Employee E
WHERE EXISTS (SELECT E2.salary FROM Employee E2
WHERE E2.DeptName = 'CS'
AND E.salary > E2.salary)
```

It takes one tuple from the Employee Relation and displays its ID if the WHERE EXISTS returns true i.e. the subquery returns one or more records. This happens in the case when the tuple from the Employee Relation E has the value of the salary attribute greater than any one of the values of the salary attribute filtered out above.
So tuples filtered out would be all the tuples that have their salary attribute value greater than the salary values of at least one from the E2 relation (3000 and 7000).
ID salary DeptName
2 40000 EC
4 40000 ME
5 50000 ME
6 60000 ME
7 70000 CS

Finally it displays their ID's and the output would be:
2
4
5
6
7
Hence option (C) 5 rows.

- d. 6

Your answer is correct.

The correct answer is: 5

Question 3

Correct

Mark 1.00 out of 1.00

Consider the following relation

Cinema (theater, address, capacity)

Which of the following options will be needed at the end of the SQL query

SELECT P1. address FROM Cinema P1

Such that it always finds the addresses of theaters with maximum capacity?

Select one or more:

- [x] a. WHERE P1. Capacity> = All (select P2. Capacity from Cinema P2)

Correct Answer is : WHERE P1. Capacity> = All (select P2. Capacity from Cinema P2)
When the **ALL** condition is followed by a list, the optimizer expands the initial condition to all elements of the list and strings them together with AND operators.
When the **ANY** condition is followed by a list, the optimizer expands the initial condition to all elements of the list and strings them together with OR operators

- [ ] b. WHERE P1. Capacity> = Any (select P2. Capacity from Cinema P2)
- [ ] c. WHERE P1. Capacity > All (select max(P2. Capacity) from Cinema P2)
- [ ] d. WHERE P1. Capacity > Any (select max (P2. Capacity) from Cinema P2)

Your answer is correct.

The correct answer is: WHERE P1. Capacity> = All (select P2. Capacity from Cinema P2)

Question 4

Correct

Mark 1.00 out of 1.00

In a schema with attributes A, B, C, D and E following set of functional dependencies are given

$A \rightarrow B$

$A \rightarrow C$

$CD \rightarrow E$

$B \rightarrow D$

$E \rightarrow A$

Which of the following functional dependencies is NOT implied by the above set?

Select one or more:

- [ ] a. $BC \rightarrow CD$
- [ ] b. $AC \rightarrow BC$
- [x] c. $BD \rightarrow CD$

Answer is (B).
Apply membership test for all the given Functional Dependencies.

i.e. BDBD cannot derive CDCD and hence is not implied.
Similarly do for rest two.

- [ ] d. $CD \rightarrow AC$

Your answer is correct.

The correct answer is: $BD \rightarrow CD$

Question 5

Correct

Mark 1.00 out of 1.00

The relation scheme Student Performance (name, courseNo, rollNo, grade) has the following functional dependencies:
name, courseNo → grade
rollNo, courseNo → grade
name → rollNo
rollNo → name
The highest normal form of this relation scheme is

Select one or more:

- a. 4NF
- b. BCNF
- c. 2NF
- d. 3NF ✓

For easy understanding let's say attributes (name, courseNo, rollNo, grade) be (A,B,C,D). Then given FDs are as follows:
AB->D, CB->D, A->C, C->A
Here there are two Candidate keys, AB and CB.
Now AB->D and CB->D satisfy BCNF as LHS is superkey in both.
But, A->C and C->A, doesn't satisfy BCNF. Hence we check for 3NF for these 2 FDs.
As C and A on RHS of both the FDs are prime attributes, they satisfy 3NF.
Hence for the whole relation the highest normal form is 3NF.

Your answer is correct.

The correct answer is: 3NF

Question 6

Correct

Mark 1.00 out of 1.00

What is the output of the following application?

```
package stocks;
public class Bond {
private static int price = 5;
public boolean sell() {
if(price<10) {
price++;
return true;
} else if(price>=10) {
return false;
}
}
public static void main(String[] cash) {
new Bond().sell();
new Bond().sell();
new Bond().sell();
System.out.print(price);
}
}
```

Select one or more:

- a. 5
- b. 6
- c. The code does not compile ✓

The sell() method does not compile because it does not return a value if both of the if-then statements' conditional expressions evaluate to false. While logically, it is true that price is either less than 10 or greater than or equal to 10, the compiler does not know that. It just knows that if both if-then statements evaluate to false, then it does not have a return value, therefore it does not compile.

- d. 8

Your answer is correct.

The correct answer is: The code does not compile

Question 7

Correct

Mark 1.00 out of 1.00

Which of the following can fill in the blank to make the class compile?

```
package ai;
public class Robot {
_____ compute()
{ return 10; }
}
```

Select one or more:

- [ ] a. private String
- [x] b. Long

Option A is incorrect because the public access modifier starts with a lowercase letter. Options C and D are incorrect because the return types, void and String, are incompatible with the method body that returns an integer value of 10. Option B is correct and has package-private access. It also uses a return type of Long that the integer value of 10 can be easily assigned to without an explicit cast.

- [ ] c. void
- [ ] d. Public int

Your answer is correct.

The correct answer is: Long

Question 8

Correct

Mark 1.00 out of 1.00

Which line of code, inserted at line p1, causes the application to print 5?

```
package games;
public class Jump {
private int rope = 1;
protected boolean outside;
public Jump() {
// p1
outside = true;
}
public Jump(int rope) {
this.rope = outside ? rope : rope+1;
}
public static void main(String[] bounce) {
System.out.print(new Jump().rope);
}
}
```

Select one or more:

- [ ] a. this(5);
- [x] b. this(4);

First off, all of the lines compile but they produce various different results. Remember that the default initialization of a boolean instance variable is false, making outside false at line p1. Therefore, this(4) will cause rope to be set to 5, while this(5) will cause rope to be set to 6. Since 5 is the number we are looking for, Option A is correct, and Option C is incorrect. Option B is incorrect. While the statement does create a new instance of Jump, with rope having a value of 5, that instance is nested and the value of rope does not affect the surrounding instance of Jump that the constructor was called in. Option D is also incorrect. The value assigned to rope is 4, not the target 5.

- [ ] c. rope = 4;
- [ ] d. new Jump(4);

Your answer is correct.

The correct answer is: this(4);

Question 9

Correct

Mark 1.00 out of 1.00

Given the following class, what should be inserted into the two blanks to ensure the class data is properly encapsulated?

```
package storage;
public class Box {
public String stuff;
_____String_____ () {
return stuff;
}
public void setStuff(String stuff) {
this.stuff = stuff;
}
}
```

Select one or more:

- [x] a. None of the above

The class data, stuff, is declared public, allowing any class to modify the stuff variable and making the implementation inherently unsafe for encapsulation. Therefore, there are no values that can be placed in the two blanks to ensure the class properly encapsulates its data, making Option D correct. Note that if stuff was declared private, Options A, B, and C would all be correct. Encapsulation does not require JavaBean syntax, just that the internal attributes are protected from outside access, which all of these sets of values do achieve.

- [ ] b. public and setStuff
- [ ] c. public and getStuff
- [ ] d. private and isStuff

Your answer is correct.

The correct answer is: None of the above

Question 10

Correct

Mark 1.00 out of 1.00

How many final modifiers would need to be removed for this application to compile?

```
package end;
public final class Games {
public final static int finish(final int score) {
final int win = 3;
final int result = score++ < 5 ? 2 : win;
return result+=win;
}
public static void main(final String[] v) {
System.out.print(finish(Integer.parseInt(v[0])));
}
}
```

Select one or more:

- [x] a. Two

The finish() method modifies two variables that are marked final, score and result. The score variable is modified by the post-increment ++ operator, while the result variable is modified by the compound addition += operator. Removing both final modifiers allows the code to compile. For this reason, Option C is the correct answer.

- [ ] b. The code will not compile regardless of the number of final modifiers that are removed.
- [ ] c. None
- [ ] d. One

Your answer is correct.

The correct answer is: Two

Question 11

Correct

Mark 1.00 out of 1.00

Which statement(s) about the following class would help to properly encapsulate the data in the class?

```
package shield;
public class Protect {
private String material;
protected int strength;
public int getStrength() {
return strength;
}
public void setStrength(int strength) {
this.strength = strength;
}
}
```

I. Change the access modifier of strength to private.
II. Add a getter method for material.
III. Add a setter method for material.

Select one or more:

- a. II and III
- b. I, II, and III
- c. I ✓

The access modifier of strength is protected, meaning subclasses and classes within the same package can modify it. Changing the value to private would improve encapsulation by making the Protect class the only one capable of directly modifying it. For these reasons, the first statement is correct. Alternatively, the second and third statements do not improve the encapsulation of the class. While having getters and setters for private variables is helpful, they are not required. Encapsulation is about protecting the data elements. With this in mind, it is clear the material variable is already protected. Therefore, Option A is the correct answer.

- d. None, the data in the class is already encapsulated.

Your answer is correct.

The correct answer is: I

Question 12

Correct

Mark 1.00 out of 1.00

Which of the following lines of code can be inserted in the line below that would allow the class to compile?

```
package farm;
public class Coop {
public final int static getNumberOfChickens() {
// INSERT CODE HERE
}
}
```

Select one or more:

- [x] a. None of the above

The code does not compile, regardless of what is inserted into the line because the method signature is invalid. The return type, int, should go before the method name and after any access, final, or static modifiers. Therefore, Option D is the correct answer. If the method was fixed, by swapping the order of int and static in the method declaration, then Option C would be the correct answer. Options A and B are still incorrect, though, since each uses a return type that cannot be implicitly converted to int.

The code does not compile, regardless of what is inserted into the line because the method signature is invalid. The return type, int, should go before the method name and after any access, final, or static modifiers. Therefore, Option D is the correct answer. If the method was fixed, by swapping the order of int and static in the method declaration, then Option C would be the correct answer. Options A and B are still incorrect, though, since each uses a return type that cannot be implicitly converted to int.
The code does not compile, regardless of what is inserted into the line because the method signature is invalid. The return type, int, should go before the method name and after any access, final, or static modifiers. Therefore, Option D is the correct answer. If the method was fixed, by swapping the order of int and static in the method declaration, then Option C would be the correct answer. Options A and B are still incorrect, though, since each uses a return type that cannot be implicitly converted to int.
The code does not compile, regardless of what is inserted into the line because the method signature is invalid. The return type, int, should go before the method name and after any access, final, or static modifiers. Therefore, Option D is the correct answer. If the method was fixed, by swapping the order of int and static in the method declaration, then Option C would be the correct answer. Options A and B are still incorrect, though, since each uses a return type that cannot be implicitly converted to int.
The code does not compile, regardless of what is inserted into the line because the method signature is invalid. The return type, int, should go before the method name and after any access, final, or static modifiers. Therefore, Option D is the correct answer. If the method was fixed, by swapping the order of int and static in the method declaration, then Option C would be the correct answer. Options A and B are still incorrect, though, since each uses a return type that cannot be implicitly converted to int.

- [ ] b. return 5L;
- [ ] c. return 10;
- [ ] d. return 3.0;

Your answer is correct.

The correct answer is: None of the above

Question 13

Correct

Mark 1.00 out of 1.00

What is a possible output of the following application?

```
package wrap;
public class Gift {
private final Object contents;
protected Object getContents() {
return contents;
}
protected void setContents(Object contents) {
this.contents = contents;
}
public void showPresent() {
System.out.print("Your gift: "+contents);
}
public static void main(String[] treats) {
Gift gift = new Gift();
gift.setContents(gift);
gift.showPresent();
}
}
```

Select one or more:

- a. It compiles but throws an exception at runtime.
- b. Your gift: Your gift:
- c. It does not compile. ✓

The code contains a compilation problem in regard to the contents instance variable. The contents instance variable is marked final, but there is a setContents() instance method that can change the value of the variable. Since these two are incompatible, the code does not compile, and Option C is correct. If the final modifier was removed from the contents variable declaration, then the expected output would be of the form shown in Option A.

- d. Your gift: wrap.Gift@29ca2745

Your answer is correct.

The correct answer is: It does not compile.

Question 14

Correct

Mark 1.00 out of 1.00

Given the following two classes, each in a different package, which line inserted below allows the second class to compile?

```
package clothes;
public class Store {
public static String getClothes() { return "dress"; }
}
package wardrobe;
// INSERT CODE HERE
public class Closet {
public void borrow() {
System.out.print("Borrowing clothes: "+getClothes());
}
}
```

Select one or more:

- a. static import clothes.Store.getClothes;
- b. import clothes.Store.*;
- c. import static clothes.Store.getClothes; ✓

Option A is incorrect because the keywords static and import are reversed. The Closet class uses the method getClothes() without a reference to the class name Store, therefore a static import is required. For this reason, Option B is incorrect since it is missing the static keyword. Option D is also incorrect since static imports are used with members of the class, not a class name. Finally, Option C is the correct answer since it properly imports the method into the class using a static import.

- d. import static clothes.Store;

Your answer is correct.

The correct answer is: import static clothes.Store.getClothes;

Question 15

Correct

Mark 1.00 out of 1.00

What is the output of the following application?

```
package pet;
public class Puppy {
public static int wag = 5; // q1
public void Puppy(int wag) { // q2
this.wag = wag;
}
public static void main(String[] tail) {
System.out.print(new Puppy(2).wag); // q3
}
}
```

Select one or more:

- a. 2
- b. It does not compile because of line q1.
- c. It does not compile because of line q2.
- d. It does not compile because of line q3. ✓

The Puppy class does not declare a constructor, so the default no-argument constructor is automatically inserted by the compiler. What looks like a constructor in the class is actually a method that has a return type of void. Therefore, the line in the main() method to create the new Puppy(2) object does not compile, since there is no constructor capable of taking an int value, making Option D the correct answer.

Your answer is correct.

The correct answer is: It does not compile because of line q3.

/